StudySapiens Board MCQs

Chapter 5 — Enzymes

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. Enzymes are best described as:

(a) Biocatalysts
(b) Hormones only
(c) Energy storage molecules
(d) Structural proteins only

2. Enzymes speed up reactions by:

(a) Lowering activation energy
(b) Being consumed in the reaction
(c) Increasing activation energy
(d) Changing the equilibrium constant

3. The protein part of an enzyme is called the:

(a) Coenzyme
(b) Prosthetic group
(c) Apoenzyme
(d) Holoenzyme

4. The complete active enzyme with its cofactor is called the:

(a) Inhibitor
(b) Holoenzyme
(c) Apoenzyme
(d) Substrate

5. The small region of an enzyme where catalysis occurs is the:

(a) Binding pocket only
(b) Nucleotide site
(c) Active site
(d) Allosteric site only

6. The active site consists of:

(a) Only lipids
(b) Only carbohydrates
(c) The whole enzyme
(d) A few amino acids

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7. The two regions of the active site are:

(a) Heavy and light chains
(b) Cis and trans faces
(c) Binding site and catalytic site
(d) A and B bands

8. Metal ions such as Zn2+ and Mg2+ often act as:

(a) Products
(b) Substrates
(c) Cofactors
(d) Inhibitors only

9. A cofactor covalently bound to an enzyme is called a:

(a) Coenzyme
(b) Prosthetic group
(c) Substrate
(d) Apoenzyme

10. Loosely attached organic cofactors are called:

(a) Inhibitors
(b) Prosthetic groups
(c) Coenzymes
(d) Apoenzymes

11. NAD+ is an example of a:

(a) Coenzyme
(b) Apoenzyme
(c) Substrate only
(d) Prosthetic group only

12. When NAD+ accepts hydrogen, it becomes:

(a) NADP+
(b) FAD
(c) NADH
(d) ATP

13. Many vitamins function as:

(a) Structural proteins
(b) DNA bases
(c) Cell wall components
(d) Coenzymes or parts of coenzymes

14. The lock-and-key model was proposed by:

(a) Emil Fischer
(b) Louis Pasteur
(c) Watson and Crick
(d) Daniel Koshland

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15. The induced fit model was proposed by:

(a) Emil Fischer
(b) Carl Woese
(c) Robert Hooke
(d) Daniel Koshland

16. According to the induced fit model, the active site is:

(a) Rigid and unchanging
(b) Absent
(c) Made of DNA
(d) Flexible and changes shape upon substrate binding

17. The enzyme-substrate complex is abbreviated as:

(a) ES complex
(b) ATP complex
(c) EP complex
(d) EI complex

18. After catalysis, the enzyme:

(a) Becomes a substrate
(b) Remains unaltered and can be reused
(c) Is converted to product
(d) Is destroyed

19. The optimum temperature for most human enzymes is about:

(a) 37°C
(b) 50°C
(c) 70°C
(d) 25°C

20. High temperature decreases enzyme activity mainly by causing:

(a) More cofactors
(b) Increased substrate concentration
(c) Lower pH only
(d) Denaturation

21. Pepsin works best in:

(a) Alkaline medium
(b) Strongly basic medium
(c) Neutral medium only
(d) Acidic medium

22. Trypsin works best in:

(a) Neutral only
(b) Very low pH
(c) Alkaline medium
(d) Strongly acidic medium

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23. Changes in pH affect enzyme activity by altering:

(a) Ribosome size
(b) Lipid composition
(c) Ionization of amino acids at the active site
(d) DNA sequence

24. If substrate is unlimited, increasing enzyme concentration will:

(a) Decrease reaction rate
(b) Denature the enzyme
(c) Have no effect
(d) Increase reaction rate

25. When all enzyme active sites are occupied, further increase in substrate will:

(a) Not increase the reaction rate
(b) Increase the rate indefinitely
(c) Stop the reaction completely
(d) Denature the enzyme

26. A chemical that blocks enzyme activity is called an:

(a) Substrate
(b) Activator
(c) Inhibitor
(d) Coenzyme

27. A competitive inhibitor resembles the:

(a) Cofactor only
(b) Product
(c) Enzyme protein only
(d) Substrate

28. Malonic acid inhibits succinic dehydrogenase by acting as a:

(a) Coenzyme
(b) Irreversible inhibitor only
(c) Competitive inhibitor
(d) Non-competitive inhibitor

29. A non-competitive inhibitor binds to:

(a) The active site only
(b) The substrate
(c) The product only
(d) A site other than the active site

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30. Reversible inhibitors usually bind through:

(a) Phosphodiester bonds
(b) Weak bonds such as hydrogen bonds
(c) Peptide bonds
(d) Covalent bonds only

31. Irreversible inhibitors usually form:

(a) Ionic bonds only
(b) Covalent bonds with the enzyme
(c) Hydrogen bonds only
(d) No bonds

32. Penicillin acts as an irreversible inhibitor of enzymes involved in:

(a) Bacterial cell wall synthesis
(b) Protein synthesis only
(c) Photosynthesis
(d) DNA replication only

33. Cyanide inhibits cellular respiration by blocking:

(a) Amylase
(b) Lipase
(c) Cytochrome oxidase
(d) Pepsin

34. Aspirin inhibits enzymes that produce:

(a) Insulin
(b) Haemoglobin
(c) Prostaglandins
(d) Cellulose

35. Feedback inhibition usually involves the end product inhibiting:

(a) The first enzyme of the pathway
(b) Ribosomes
(c) All enzymes equally
(d) The last enzyme only

36. Oxidoreductases catalyse:

(a) Oxidation-reduction reactions
(b) Isomerization only
(c) Hydrolysis only
(d) Group transfer only

37. Transferases catalyse:

(a) Oxidation only
(b) Transfer of functional groups
(c) Joining with ATP only
(d) Hydrolysis

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38. Hydrolases catalyse:

(a) Reduction only
(b) Oxidation only
(c) Isomerization only
(d) Hydrolysis reactions

39. Lyases catalyse:

(a) Group transfer only
(b) Non-hydrolytic addition or removal of groups
(c) Hydrolysis only
(d) Oxidation only

40. Isomerases catalyse:

(a) Intramolecular rearrangement
(b) Oxidation
(c) Polymerization
(d) Hydrolysis

41. Ligases catalyse:

(a) Isomerization only
(b) Oxidation only
(c) Hydrolysis only
(d) Joining of two molecules using ATP

42. Hexokinase transfers a phosphate group from ATP to glucose; it is a:

(a) Ligase
(b) Hydrolase
(c) Lyase
(d) Transferase

43. Pyruvate decarboxylase removes CO_2 from pyruvate; it is a:

(a) Lyase
(b) Hydrolase
(c) Oxidoreductase
(d) Isomerase

44. Digestive enzymes such as amylase and lipase are:

(a) Isomerases
(b) Hydrolases
(c) Transferases only
(d) Ligases

45. Proteases act on:

(a) Proteins
(b) Carbohydrates
(c) Nucleic acids
(d) Lipids

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46. Lipases act on:

(a) Lipids
(b) Carbohydrates
(c) Proteins
(d) DNA

47. Amylase acts on:

(a) Lipid
(b) Starch
(c) Protein
(d) DNA

48. Nucleases act on:

(a) Nucleic acids
(b) Lipids
(c) Proteins
(d) Polysaccharides only

49. Pepsin is secreted in inactive form as:

(a) Pepsinogen
(b) Procarboxypeptidase
(c) Trypsinogen
(d) Chymotrypsin

50. Enzymes of glycolysis are found mainly in the:

(a) Lysosome
(b) Cytoplasm
(c) Nucleus
(d) Mitochondrial matrix

51. Enzymes of the Krebs cycle are found mainly in the:

(a) Cytoplasm
(b) Nucleus
(c) Mitochondrial matrix
(d) Thylakoid

52. The specificity of an enzyme is mainly due to the shape of its:

(a) Amino terminal only
(b) Molecular weight
(c) Active site
(d) Carboxyl terminal only

53. Which factor can denature enzymes?

(a) Low substrate concentration
(b) Optimum pH
(c) Extreme temperature or pH
(d) Presence of cofactor

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54. Thermophilic bacteria have enzymes that:

(a) Lack active sites
(b) Function at high temperatures
(c) Work only at 0°C
(d) Are always inactive

55. The rate of an enzyme-catalysed reaction may be increased by a factor of:

(a) 10 to 20
(b) 10^6 to 10^12
(c) 2 to 3
(d) 100 only

56. Which of the following is NOT a cofactor?

(a) Glucose
(b) Zn2+
(c) NAD+
(d) Mg2+

57. The binding site of an enzyme holds the substrate by:

(a) Peptide bonds
(b) Phosphodiester bonds
(c) Covalent bonds only
(d) Weak chemical forces

58. In a metabolic pathway, the product of one enzyme usually becomes:

(a) The substrate for the next enzyme
(b) A vitamin
(c) An inhibitor of all enzymes
(d) A cofactor

59. Which enzyme class would catalyse the conversion of glucose to fructose?

(a) Isomerase
(b) Oxidoreductase
(c) Hydrolase
(d) Ligase

60. Which enzyme class joins DNA nucleotides during replication?

(a) Ligase/polymerase-related joining
(b) Hydrolase only
(c) Lyase only
(d) Isomerase only

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61. Competitive inhibition can often be overcome by:

(a) Increasing temperature only
(b) Adding more inhibitor
(c) Removing cofactor
(d) Increasing substrate concentration

62. Non-competitive inhibition cannot be overcome simply by:

(a) Adding cofactor
(b) Increasing substrate concentration
(c) Adding enzyme
(d) Lowering pH

63. The optimum pH of pancreatic amylase is approximately:

(a) 3.0
(b) 1.5–2.0
(c) 10.0
(d) 6.7–7.0

64. The optimum pH of arginase is approximately:

(a) 6.0
(b) 10.0
(c) 2.0
(d) 4.0

65. Which of the following is a reversible inhibitor?

(a) Cyanide
(b) Penicillin
(c) Heavy metal covalent binders only
(d) Malonate

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ANSWER KEY WITH BRIEF REASONS

1. (a) Biocatalysts — Enzymes speed up reactions without being consumed.

2. (a) Lowering activation energy — They reduce the energy barrier.

3. (c) Apoenzyme — Apoenzyme is the protein portion.

4. (b) Holoenzyme — Holoenzyme = apoenzyme + cofactor.

5. (c) Active site — The active site binds substrate and catalyses the reaction.

6. (d) A few amino acids — Only a small number of residues form the active site.

7. (c) Binding site and catalytic site — Binding holds substrate; catalytic site performs chemistry.

8. (c) Cofactors — Metal ions assist enzyme activity.

9. (b) Prosthetic group — Prosthetic groups are tightly/covalently attached.

10. (c) Coenzymes — Coenzymes assist catalysis and may dissociate.

11. (a) Coenzyme — NAD+ carries hydrogen/electrons.

12. (c) NADH — Reduction of NAD+ forms NADH.

13. (d) Coenzymes or parts of coenzymes — B-vitamins often form coenzymes.

14. (a) Emil Fischer — Fischer proposed the rigid active site model.

15. (d) Daniel Koshland — Koshland proposed flexible active site adjustment.

16. (d) Flexible and changes shape upon substrate binding — Substrate binding induces conformational change.

17. (a) ES complex — Enzyme + substrate forms ES.

18. (b) Remains unaltered and can be reused — Enzymes are not consumed.

19. (a) 37°C — Human enzymes work best near body temperature.

20. (d) Denaturation — Heat disrupts protein structure.

21. (d) Acidic medium — Pepsin has a low optimum pH.

22. (c) Alkaline medium — Trypsin acts in the small intestine.

34. (c) Prostaglandins — Aspirin reduces inflammation/pain.

35. (a) The first enzyme of the pathway — This regulates metabolic pathways.

36. (a) Oxidation-reduction reactions — They transfer electrons/hydrogen.

37. (b) Transfer of functional groups — They move groups between molecules.

38. (d) Hydrolysis reactions — They break bonds using water.

39. (b) Non-hydrolytic addition or removal of groups — They remove/add groups without water.

40. (a) Intramolecular rearrangement — They convert isomers.

41. (d) Joining of two molecules using ATP — Ligases form new bonds with energy.

42. (d) Transferase — It transfers phosphate.

43. (a) Lyase — Decarboxylation is non-hydrolytic removal.

44. (b) Hydrolases — They hydrolyse food molecules.

45. (a) Proteins — Proteases digest proteins.

46. (a) Lipids — Lipases hydrolyse fats.

47. (b) Starch — Amylase digests starch.

48. (a) Nucleic acids — Nucleases digest DNA/RNA.

49. (a) Pepsinogen — Inactive secretion protects the stomach.

50. (b) Cytoplasm — Glycolysis occurs in cytosol.

51. (c) Mitochondrial matrix — Matrix enzymes catalyse the Krebs cycle.

52. (c) Active site — Active site geometry determines substrate fit.

53. (c) Extreme temperature or pH — Denaturation disrupts structure.

54. (b) Function at high temperatures — Their proteins are heat-stable.

55. (b) 10^6 to 10^12 — Enzymes greatly accelerate reactions.

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23. (c) Ionization of amino acids at the active site — pH changes charge and shape.

24. (d) Increase reaction rate — More active sites are available.

25. (a) Not increase the reaction rate — The enzyme is saturated.

26. (c) Inhibitor — Inhibitors reduce enzyme activity.

27. (d) Substrate — It competes for the active site.

28. (c) Competitive inhibitor — Malonate resembles succinate.

29. (d) A site other than the active site — It changes enzyme shape.

30. (b) Weak bonds such as hydrogen bonds — Weak binding can be reversed.

31. (b) Covalent bonds with the enzyme — They permanently disable the enzyme.

32. (a) Bacterial cell wall synthesis — Penicillin blocks cell wall formation.

33. (c) Cytochrome oxidase — Cyanide binds cytochrome oxidase.

56. (a) Glucose — Glucose is a substrate/fuel, not a typical cofactor.

57. (d) Weak chemical forces — Hydrogen/ionic interactions hold substrate.

58. (a) The substrate for the next enzyme — Pathways are sequential.

59. (a) Isomerase — Glucose-fructose conversion is isomerization.

60. (a) Ligase/polymerase-related joining — Polymerases join nucleotides using energy.

61. (d) Increasing substrate concentration — More substrate outcompetes inhibitor.

62. (b) Increasing substrate concentration — The inhibitor acts away from the active site.

63. (d) 6.7–7.0 — Pancreatic amylase works near neutral pH in the duodenum.

64. (b) 10.0 — Arginase has a high optimum pH.

65. (d) Malonate — Malonate inhibition can be reversed.

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